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Effect of thermal pretreatment and nanobubble addition water on methane production from cowpea hull anaerobic digestion

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Abstract

Residual cowpea hull (CH) from akara (bean cake) processing constitutes a growing organic waste in countries with high production rates. CH has an unpleasant smell but rich in carbon, nitrogen, and micronutrients, which makes it a potential feedstock for energy recovery via anaerobic digestion (AD). However, the effect of the high antinutrient content, such as trypsin inhibitor (TI) previously reported in CH, and other parameters like moisture content on methane production need to be evaluated to promote its utilization as an AD feedstock. Anaerobic co-digestion of thermally pretreated cowpea hull and waste-activated sludge was investigated in this study, with a focus on the enhancing effect of thermal pretreatment (TP) at 83 and 100% moisture content reduction (MCR) and nanobubble water (air and N_2-NBW) addition on methane production. TP of CH under 100% MCR increased methane production by 12.9% compared to fresh CH. Additionally, replacing deionized water with N_2-NBW further enhanced methane production by 20.9%. However, the effect of Air-NBW on methanogenesis was not significant. Improved CH biodegradability and the suppression of TI by TP may have facilitated the F/M interaction rate. However, the mechanisms of TI on microbial activity in an anaerobic digester need to be clarified. The electric power generating potential from the methane obtained in this

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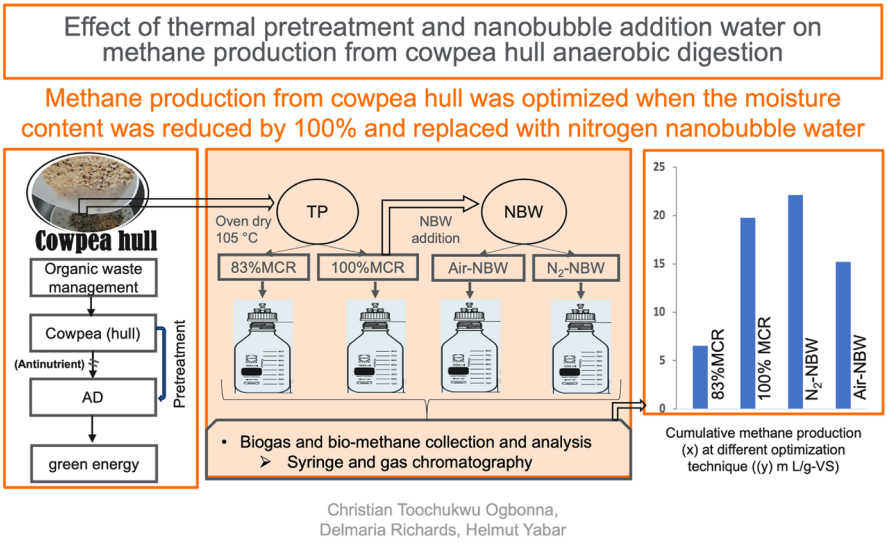
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study was also estimated to promote sustainable disposal of CH as a waste management technique and green energy recovery.

Graphical abstract



Keywords Organic waste · Cowpea hull · Moisture content reduction · Nanobubble water · Anaerobic digestion · Methane production

Introduction

Cowpea hull (CH) is becoming a critical environmental problem in the producing countries due to its high production rate and poor disposal practices [1, 2]. It is the seed coat of cowpea seed extract during cowpea/bean cake processing. The aggregate global production of cowpeas in 2019 was 9.8 million tons (MT) [3], representing a 2.7-fold expansion since 2000. Nigeria and Niger account for 80% of global cowpea production. In 2011 and 2021, Nigeria produced 1.64 and 3.62 MT of cowpea seeds [4, 5]. The country's growing waste management challenge has been extensively discussed [6], of which cowpea waste constitutes a significant portion [7]. Anaerobic digestion (AD) stands out among other organic waste treatment technologies due to its relative cost-effectiveness, clean energy source, and other nutrient recovery [8–10].

AD of CH is a potential technique for organic waste disposal and biogas production due to its wide availability in several countries [1, 11, 12] and the rich carbon and nitrogen content (18 to 25%), which is essential for AD system [13, 14]. Still, co-digestion with other nitrogen-rich substrates, such as waste-activated sludge (WAS), promotes process stability and nutritional balance [15] compared

to mono-digestion. Also, drying is imperative for CH AD to reduce its volume and weight for easy storage and transportation to the digester facility, it also improves the stinking odor and overall sanitation, especially among the rural population. Currently, limited studies have investigated the bio-gasification of CH. Biogas generation from cow dung, cowpea, and cassava peeling has been was compared, and 67.9, 56.2, and 51.4% of methane content was obtained from each feedstock, respectively [16]. Recently, Co-digestion of CH with either WAS, or rice straw (RS) showed an increased total methane production by 18.95% or 27.05%, correspondingly [2]. CH has also been investigated for glucose production via a two-stage dilute-acid hydrolysis [7].

Various pretreatment options, such as mechanical, thermal, and biochemical, could enhance the AD of organic waste, such as CH. Several studies have revealed a positive effect of thermal pretreatment (TP) on the chemical composition of feedstock and biogas generation [17–19]. For example, TP of 180 °C on wheat straw observed a 53% increase in biogas generation [17]. The mechanism for this effect is attributed to enhanced substrate solubility, which increases anaerobic biodegradability [20]. Moreover, several studies on CH for animal feed shows that it contains antinutritional factors such as trypsin inhibitors (TI) that counteract biochemical reactions [12, 21, 22]. In addition to improving solubilization, TP is an effective technique to inactivate TI in legumes, breaking the intermolecular bonds in the complex structure of the TI [22, 23]. Also, combining thermal treatment with chemical addition could increase the efficiency of TI deactivation; sodium chloride (NaCl) has enhanced the deactivation of TI in soymilk by 74% [22]. However, chemical additions are costly and constitute a secondary pollution [24]. As a non-chemical-based catalyst, nanobubble water (NBW) has unique properties that enhance biogas production in AD. It is an environmentally friendly, cost-effective technique for enhancing AD [25–28]. While past studies suggest that CH is feasible for biogas production, little is known about the influence of CH moisture content nor the effect of TI content on CH AD. Exploring these factors motivated this study. In other words, promoting CH as an attractive feedstock for biogas production requires addressing possible bottlenecks that could impede its practical application for biogas production and optimizing the fermentation process for higher biogas production comparable to the conventional AD feedstock.

This study proposes a novel strategy to enhance methane production from CH. The hypothesis is that reducing the moisture content in CH via TP correlates with an enhanced AD process due to a reduced antinutrient content in CH. Second, restoring the dividend with NBW in place of deionized water (DW) would further enhance the hydrolysis process of CH. Therefore, the effect of different percentages of moisture content reduction (MCR) on biogas production and total solid reduction from CH co-digested with WAS was investigated, followed by the effect of adding different types of NBW (air- and nitrogen-NBW) in recompensing the removed organics from moisture reduction. Finally, the study estimates the potential electricity-generating capacity from the obtained methane production. To our knowledge, systematic studies on the effect of TP and NBW addition on CH anaerobic co-digestion are yet to be investigated in the literature. The findings will reveal the new technique to enhance

the AD of CH for optimal biogas production and effective disposal methods for organic waste.

Materials and methods

Material collection and preparation

Cowpea seeds used in this study were purchased from a grocery shop in Tokyo, Japan. They were soaked in water at room temperature for 24 h to soften. The tender seeds were then manually peeled to separate the hull. The hull was dried under sunlight and sealed in a plastic bag for the experiments. Digestate sludge was used as inoculum due to the high nitrogen and bacteria content. Digestate sludge and WAS were fetched from the Shimodate Wastewater Treatment Plant in Ibaraki Prefecture, Japan. To reduce the high liquid content, it was centrifuged at 4 °C, 9000 rpm for 5 min, and stored at 4 °C before use for the batch experiment.

Thermal pretreatment and nanobubble water generation

The TP of CH was conducted in a laboratory drying oven (Yamato DX601) at 105 °C temperature. Two MCRs (83.42 and 93.85) achieved during different retention times (3 and 48 h) were tested. NB. The two conditions were hereafter referred to as 83 and 100%MCR. The samples were cooled to room temperature, ground, and filtered through a 1 mm filter dish to increase the surface area and enhance the fermentation rate. NBW was generated using an ultra-fine bubble generator (HACK UFB, Japan), as shown in Fig. 1a, is the nanobubble generator. Air or N_2 gas was injected into the bubble generator via the gas inlet (1), and 0.5 L of deionized water was connected to the water inlet (2). The pressure of the nanobubble generator was set at 0.24 MPa while gas and water flow rates were regulated at 0.6 L/h and 3 L/min

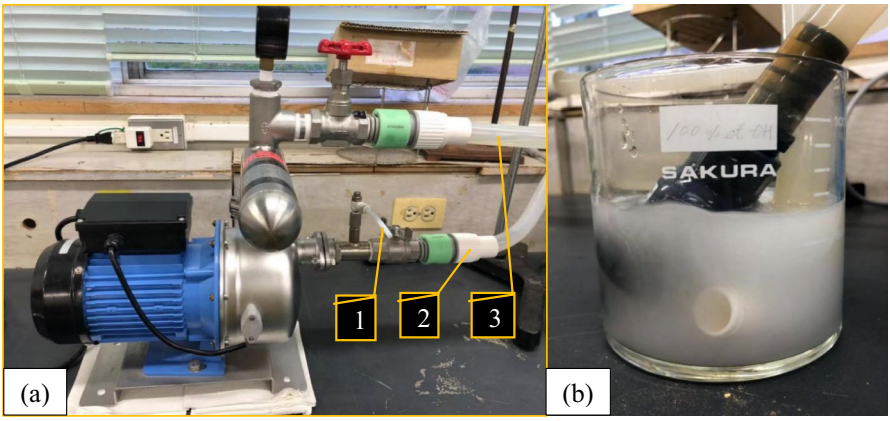


Fig. 1 Photo illustration of NB generator (**a**) and generated NBW (**b**)

Table 1 Experimental conditions for thermal pretreatment and NBW addition

Scenario	CH (g)	WAS (g)	Inoculum (g)	Final volume (mL)
Untreated CH	1.50	0.50	1	60
83% MCR	1.50	0.50	1	60
100% MCR	1.50	0.50	1	60
Air-NBW	1.50	0.50	1	60
N_2-NBW	1.50	0.50	1	60

CH is cowpea hull, MCR is moisture content reduction, NBW is nanobubble water, and WAS is waste-activated sludge. CH, WAS, and inoculum percentages were calculated based on the wet weight; VS was between 2.8–3.3 g for each test sample

Table 2 Effects of pretreatment on biogas and methane production

Scenario	Cumulative biogas (mL/g VS)	Daily CH_4 peak (mL/day)	Cumulative CH_4 (ML/g VS_{added})
Untreated CH	197.93	21.35 $^{(day\ 4)}$	104.26
N_2-NBW	242.70	31.21 $^{(day\ 3)}$	129.33

simultaneously. The retention time was limited to 25 min compared to the 30 min reported in previous studies [27, 29] due to observed generator overheating. The generated NBW was collected via the gas outlet (3) and allowed to cool at room temperature (Fig. 1b) before being transferred into the reactor.

Batch anaerobic digestion setup

Two separate batch experiments were conducted to investigate the effects of TP and NBW addition on biogas production and total solid degradation. First, 1.5 g of CH was mixed with 0.5 g of WAS, a 3:1 optimum ratio obtained from our previous study [2]. The inoculum (digestate sewage sludge) was 1 g, all based on the VS content. The first batch experiment used CH at three conditions, as indicated in Table 1. Sample 1 had fresh CH as a control experiment (Untreated CH). In samples 2 and 3, CH at 83% MCR (83%MCR) and 100% (100%MCR) were added, respectively. Deionized water (DW) was added to all samples to bring the working volume to 60 mL. In the second batch, a replicate of the first batch, sample 3, was set up as the control experiment in sample 1. In samples 2 and 3, air-NBW (Air-NBW) or nitrogen-NBW (N_2-NBW) was used to top the working volume instead of DW. All batch samples were fed into 100 mL serum glass bottles. The pH was carefully adjusted to 7 ± 05 using a 2 M NaOH or HCL. The reactors were sealed with a silica gel stopper, and nitrogen gas was injected into the reactor for 2 min to purge air and ensure an anaerobic working condition before being placed into a mesophilic incubator at 38 ± 1 °C. The experiments were conducted in duplicates, and the average results were reported (Table 2).

Analytical methods

The substrates were analyzed by the total solid (TS), volatile solid (VS), and moisture content according to the standard method [30]. For the total organic carbon (TOC), nitrogen (TN), and Hydrogen (H) content, 3 g of dried CH were ground filtered and sent to the elemental analysis center (University of Tsukuba, Japan). The dividend of TOC and TN gave the C/N ratio, while pH was measured with a pH meter (Mettler Toledo). For the cumulative biogas, gas volume was recorded every 2 days by inserting a pressure-tight glass syringe into the reactor via the soft silicate cap. Afterward, 1 mL of the gas was analyzed in gas chromatography (GC-8A, Shimadzu, Japan) equipped with a thermal conductivity detector (80 °C) and Porapak Q column (60 °C) and N_2 as carrier gas for the methane, carbon dioxide, and hydrogen content. The methane production was obtained by multiplying methane content by the total gas produced [2]. Although the inoculum's gas output was not accounted for in the net gas production, it had been cultivated for 14 days, and gas was released before use.

Quantification of potential electrical energy

Nigeria and Niger Republic produce over 80% of global cowpea production annually [31]. Approximately 3.6 million tons of cowpea are consumed annually in Nigeria, mainly for preparing bean flour and bean cake dishes such as moi-moi and akara [32]. The study acknowledges that abiotic and biotic stresses, cultivation methods, and socioeconomic constraints influence cowpea production. These limitations can impact the overall availability of CH as a potential feedstock. Consequently, these factors must be considered in understanding the feasibility of CH's production of biogas [11, 33]. However, the abundant nature of WAS and NBW is assumed to be available for co-digestion with CH. Therefore, it is contingent on sludge production from municipal and industrial sewage plants and NBW commercially produced from a nanobubble generator. The calculation utilizes Nigeria's 2020 output of 3.65 MT cowpea gross production as a case study to estimate the electrical energy generation potential. The results are calculated by quantifying the crop residue generation, collection factor, surplus availability, residue dryness factor, the ratio of volatile solids, and biogas generation rate [34]. After dehulling, 1 kg of cowpea seed gave 124.20 g of CH used in the computation for waste-to-energy reprocessing. Equations 1 and 2 determine the potential electrical energy from fresh and pretreated CH AD.

$$A_{\mathrm{CPHB}} = \sum \left(\frac{\mathrm{CPHB}}{1.5} \right) \mathrm{xBGRxMEC} \tag{1}$$

where: A_{CPHB} is the available CH for biogas, CPHB represents CH for biogas in grams, 1.5 is the weight of the CH in grams utilized in the biogas experiment, BGR is equivalent to the average biogas generation rate per day in m^3, and MEC is the methane energy content based on volatile solids generated in the biogas process in percentage.

$$\mathrm{PE}_{\mathrm{CHPB}} = \frac{(A_{\mathrm{CPHB}} \mathrm{x2kWh})}{1,000,000} \quad (2)$$

where: $\mathrm{PE}_{\mathrm{CHPB}}$ is potential electrical energy from biogas in m^3, which is then expressed in kWh, assuming 1 m^3 is equivalent to 2 kWh.

Results and discussion

The effect of CH at two different TP (83 and 100% MCR) and two different NBW addition (air and N_2-NBW) was investigated in wet mesophilic anaerobic co-digestion with WAS for optimizing biogas production and total solid degradation. The initial pH was 7.0, and the working temperature was controlled at 38 ℃. The AD duration was 16 days.

Characteristics of the feedstock

The physicochemical characteristics of feedstock are determined for their possible influence on the stability and efficiency in the batch experiment. Table 3 shows the characteristics of CH, WAS, and inoculum used in this study. Notably, CH has a total solid content of 92%, of which volatile solid accounts for 96.45%, implying a highly degradable portion that could be exploited for biogas production. Although WAS has a lower TS (48%), the volatile solids per TS is also high (95.46%). The high nitrogen content in WAS (6.8%) has been used to compensate for the low nitrogen content in rice straw (0.9%) co-digestion [35]. The carbon content in CH is 44.84 wt%, while the nitrogen content is 2.2 wt%, implying a C/N ratio of 20.02. Compared to 1.5 wt% of nitrogen in cowpea seed [36], CH is more rich in nitrogen.

Effect of thermal pretreatment on biogas and methane production

The effect of TP was investigated to enhance the efficiency of AD of CH and WAS using digestate sludge as inoculum. TP is expected to improve the solubility of CH, breaking the rigid cell wall structure that prevents organic matter digestibility during the AD of lignocellulosic materials. In doing so, the activity of the high antinutrient content in CH, such as trypsin inhibitor (TI), could equally be reduced. The

Table 3 Characteristics of cowpea hull, waste-activated sludge, and inoculum

Parameters	CH	WAS	Inoculum
pH	ND	6.09	6.98
TS (%)	92.90	48.50	10.20
VS (%)	96.45	95.46	81.37
CH elemental composition: TOC (wt%) 44.84, TN (wt%) 2.24, C/N ratio 20.02			

CH—Cowpea hull; WAS—waste-activated sludge; TS—total solids; VS—Volatile solids

efficiency of AD in this study is evaluated by the increase in the cumulative biogas, methane production and the total solid degradation (discussed separately). Figure 2 shows the biogas production from the co-digestion of CH.

A significant increase in biogas production was achieved by TP at 100% MCR, i.e., 219.14 mL/g VS, followed by 83% MCR (217.02 mL/g VS). Compared to fresh CH (197.93 mL/g VS), this is a 10.7 and 9.6% increase in total biogas production. The result implies that increasing the MCR in CH achieved by the longer retention time of TP resulted in enhanced anaerobic digestion. The daily biogas trend shows that all samples produced an equal amount of biogas on the first day (Fig. 2). However, the untreated CH had accumulated more biogas on day 2 (70 mL) compared to 100% MCR (60 mL). By day 4, the CH at 100% MCR and 83% MCR exceeded the untreated CH, and the trend was sustained until the end of fermentation on day 16, when biogas production dropped to less than 1.0 mL/day. The cumulative methane production followed a similar trend as the biogas production. TP at 83 and 100%MCR produced more methane at 116.15 and 114.41 mL/g VS, respectively, compared to fresh CH (104.26 mL/g VS).

TP improves the substrate solubilization in an AD system [20]. It disintegrates cell walls and facilitates the release of intracellular matter into the aqueous phase, which augments biodegradability, thus improving AD with lower retention time and higher biogas generation [37, 38]. It also increases WAS AD's microbial populations and biogas yield, yielding the highest biogas rate of 0.42 L/g$-^{1}$ volatile solid (VS) [18]. Although the result obtained in this study is significant, the increase in biogas and methane production due to TP was lower than those reported in the literature utilizing the same pretreatment method. For example, Coffee husk TP at 120 and 180 °C resulted in a 7.64 and 8.98 g TOC/L increase in organic compounds solubilization, respectively [20]. Pineapple peel waste in a batch AD system showed enhanced biogas generation and decreased AD process lag time, resulting in the highest biogas production volume of 616.33 mL (357.190 mL/g VS) [19].

Regarding the more biogas production from fresh CH at the start of the fermentation (Fig. 2), this could be attributed to the high galactooligosaccharides content in

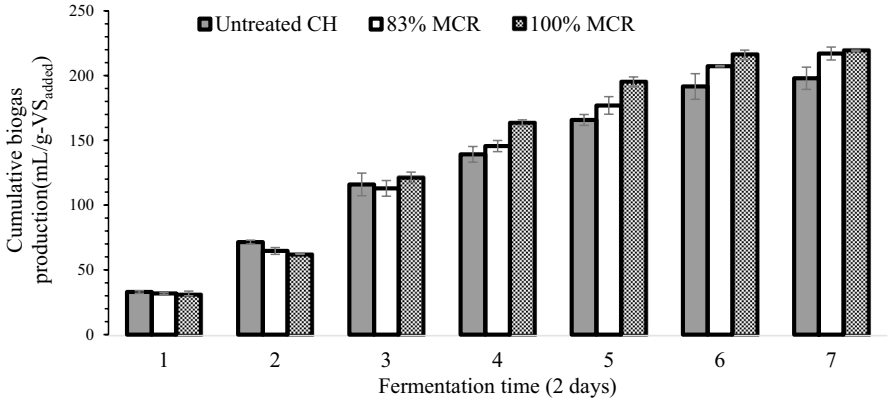


Fig. 2 Effect of thermal pretreatment on cumulative biogas production

CH, which exhibits quick fermentation under anaerobic conditions [39]. However, the biogas potential was unsustainable and steadily decreased when other antinutrients began to degrade in subsequent days. Dehulling cowpea seeds has shown a 30.25% and 26.80% reduction of raffinose and stachyose (galactooligosaccharides types) [40], confirming that a significant portion of both probiotics and antinutrients remains on CH. Although the significant enhancement in substrate AD degradation, evident in the biogas and methane production, could be attributed to the increased solubility. Several studies on CH's nutritional characteristics for animal feed indicate a high content of antinutrients such as TI. A total of 7.52 U/mg of TI activity in raw cowpea has been reported [41]. Trypsin enhances methane production, facilitating hydrolysis and acetic acid conversion during the AD of WAS [42]. On the contrary, TI reduces the activation and catabolism of proteins [43], including the digestion of proteins [22]. It is assumed in this study that at no pretreatment, fresh CH having a high content of TI resulted in a decrease in methane production due to repressed microbial activities in the AD system. Hence, microbial analysis of CH before and after TP is recommended as a prerequisite step in further study on CH AD to ascertain the effect of TI.

Effect of nanobubble water addition on biogas and methane production

In the second batch experiment, the Cowpea hull at 100 percent MCR was used to investigate the synergistic effect of TP and NBW addition on CH AD. Figure 3 shows the effect of different NBW types (Air and nitrogen) on daily biogas and cumulative methane production after 16 days of batch retention time. The daily biogas production (Fig. 3a) shows that biogas production was over 30 mL/day in all setups on day 1, while the peak biogas production was recorded on day 3. The biogas production on day 3 was 59.06, 60.11, and 64.51 for CH at 100% MCR, Air-NBW, and N_2-NBW addition. On the other hand, the cumulative methane production was 116.15, 116.83, and 129 mL/g VS for CH at 100% MCR, Air-NBW, and N_2-NBW

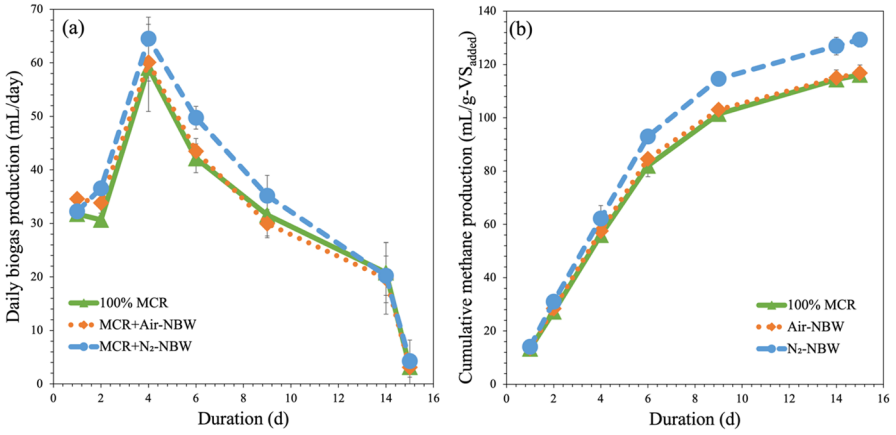


Fig. 3 Cumulative biogas (**a**) and daily methane production (**b**) during AD after air or N_2-NBW addition

addition, respectively (Fig. 3b). Nitrogen-NBW significantly enhanced methane production by 11.35% compared to CH without NBW. However, methane production was only slightly improved (0.59%) when air-NBW was used. While N_2-NBW had a relatively similar enhancement in biogas production (10.5%), Air-NBW, on the other hand, enhanced total biogas production by threefold (2.5%) compared to methane production.

Microbial activities in water treatment and fermentation of human excrement have been enhanced with NBW addition [26]. Although significant, the enhancing effect of NBW recorded in this study is relatively low compared to previous studies. A 29% increase in methane production was achieved in WAS single digestion when N_2-NBW was added, with no noticeable change in methane content. [27]. A 39% increase in cumulative methane production was obtained when N_2-NBW was added to pig manure due to enhanced hydrolysis [44]. Similarly, a follow-up investigation showed a 25.0% increase from the same substrate under high solids (3–6% TS)[45]. The highest cumulative methane production (132.16 mL g VS) was achieved when NBW was added to corn straw AD [28]. Nitrogen-NBW promoted seed germination rate by 25% compared to water, while no significant effect was seen with air or carbon dioxide-NBW [46]. Methanogenic activity is sensitive to different environments [47] and may have been altered by antinutrients in CH that could not be removed entirely by TP.

Another assumption for the relatively low enhancing effect of NBW in this study is that NBW may have rapidly decomposed in the reactor during the 16 days of fermentation. On the contrary, a study showed that N_2, O_2, Ar, and air-NBW prepared under an aqueous solution lasted over 60 days [48]. However, they also found that CO_2 was unstable in water and lasted only 5 days [48]. NBW used in this study may not have been fully formed due to technical errors during the bubble generation, leading to short stability in the reactor. Recall that the retention time for the NBW was 5 min less than the recommended 30 min in previous studies [27, 29]. This assumption also explains the lower enhancement from air-NBW compared to N_2-NBW, as decomposed air-NBW will rapidly oxidize the anaerobic working condition compared to N_2-NBW.

Overall, our findings revealed that NBW has a different effect on hydrolysis and methanogenesis based on the type of NBW and the reacting substrate. However, there tends to be a consensus on the positive enhancing effect of NBW on AD systems in the literature. This study, therefore, recommends analyzing the characteristics of the generated NBW, such as the mobility and zeta potential, before the AD batch experiment to ascertain the stability of the bubbles. And at the same time, taking a careful consideration of the targeted substrates characteristics.

While successful optimization of AD following the direct addition of different chemicals and complex enzymes such as carbohydrolases, peptidases, and lipases has been reported [37]. NBW comprises cheap gasses with the potential to improve the anaerobic condition of the methanogen [29]. NBW had a positive effect in this study, particularly nitrogen-NBW enhanced hydrolysis, and methanogenesis in the AD of CH. Requiring only the initial cost for the bubble generator and the inexpensive nitrogen and air gases, NBW is a relatively cheap and environmentally friendly technology compared to other optimization options. Hence, NBW-supplemented AD

of organic waste is recommended for waste management and green energy recovery, especially in developing countries where the cost of operation and complex technical know-how are of major concern.

The pH trend and total organic solid decomposition

After 16 days of mesophilic anaerobic digestion with initial pH adjusted to 7 ± 05. The pH trend of all the samples remained within 6.8–7.0. Fresh CH, 83% MCR, 100% MCR, Air-NBW, and N_2-NBW had a final pH values of 6.90, 6.87, 6.85, 6.84, and 6.89, respectively (Fig. 4), affirming that methanogens generally survive in a neutral pH (7) conditions to biodegrade the feedstock [49]. The most stable pH trend was found in N_2-NBW, having the lowest acidification rate (6.84), Similarly, Hesborn et al. [50] achieved the same pH trend (6.8) when the initial alkalinity of the digester was increased by 6000 g/L compared to unadjusted alkalinity. On the contrary, another study found that acid formers could be best controlled at a pH of 6 for the bioconversion of citrus waste [51]. Notwithstanding, the relatively stable pH in all reactors conforms with the successful fermentation of CH and WAS within 16 days of batch retention time, achieving a high average methane percentage (70%) in all test groups. A customarily operated AD system produces biogas with methane content ranging from 55 to 70% [52]. This implies that optimum working pH condition was maintained [53].

The total and volatile solid reduction is not only an effective indicator of the efficiency of the AD performance in this study. It is considered essential as an organic waste disposal technique. Figure 5 shows the total solid and volatile solid reduction after the AD. All seven batch setups had a similar total solid and volatile solid reduction trend. The total solid reduction was 34% in all reactors except for Air-NBW, which had a 32.5% TS reduction. Correspondingly, the volatile solid reduction was 44% in Air-NBW, 45% with no NBW addition (100% MCR), and slightly above 45% in other batch reactors. Although lignins are difficult to degrade, our

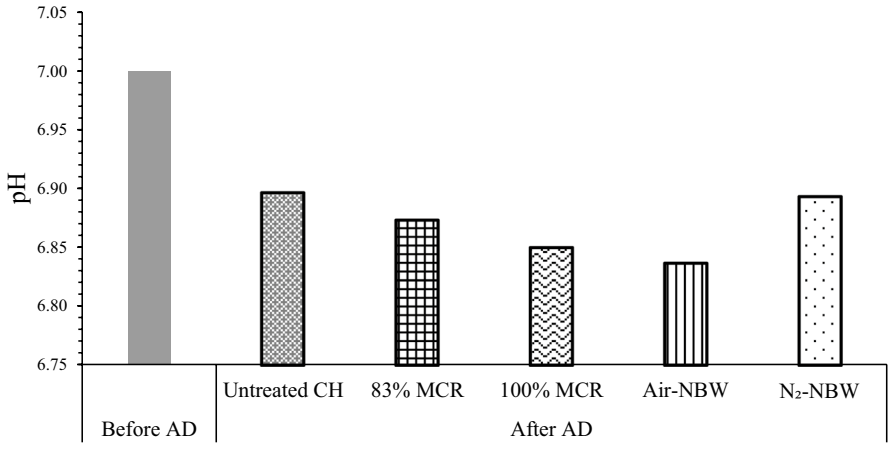


Fig. 4 Average pH variation after AD at different pretreatment condition

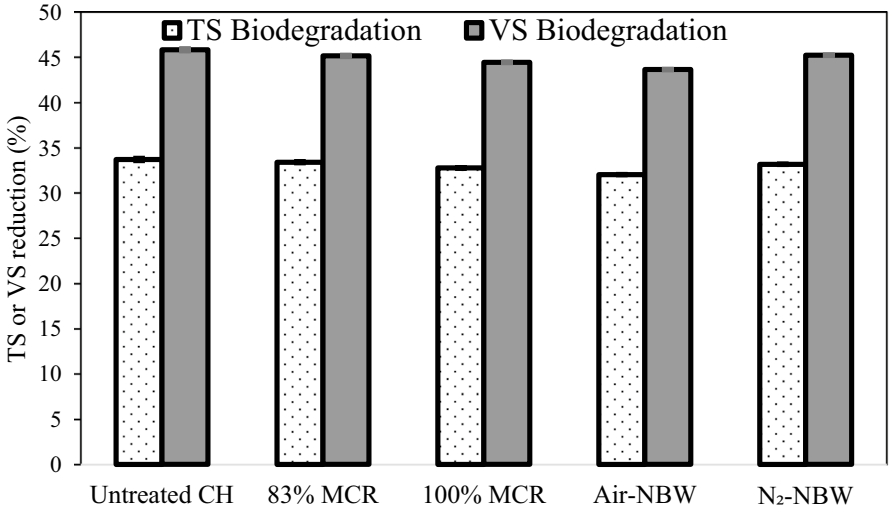


Fig. 5 Total solid and volatile solid reduction after AD at different pretreatment

result is much higher than the 17.41% or 19.08% solid reduction obtained when CO_2 or N_2-NBW were added to lignin AD, respectively [29]. The result is consistent with the suggested 40–50% volatile solid reduction under normal AD working conditions [54]. The low solid reduction rate when Air-NBW was added corresponds with the low methane produced in the Air-NBW setup. This suggests that the hydrolysis was not enhanced, possibly due to unstable nanobubbles that rapidly degrade and alter the proper anaerobic working condition.

Electrical energy generation

The theoretical electrical energy potential is derived from feedstock utilized in the experiment and computation contingent on Nigeria's production of CH and decorticating for food preparation. Thus, the results are amenable. The computation used the average daily methane production (Table 3). The result reveals that the potential biogas of 47.13 and 68.90 million m^3 generated by co-fermentation of untreated and thermally treated CH with WAS under anaerobic conditions can be converted to 50.96 − 75.36 GWh. Furthermore, it addresses the ability to reduce carbon-based electrical energy, which constituted 77% of the 27,042 GWh electricity generated in 2020 [55].

Consequently, 0.19 to 0.27% of the total electricity capacity generated in 2020 could be replaced by the energy recovered from waste substrates utilized in the experiment. Besides, biogas could be used as cooking fuel, especially in Nigeria's remote areas, where households depend primarily on traditional biomass and kerosene [56]. The green energy generated can aid in mitigating methane emissions by recycling organic waste that would otherwise escape to the atmosphere from open dump landfilling, which is common in Nigeria [57]. The synergistic benefit of AD

technology has been carefully estimated using the case study of Yuge Farm in Japan [58] and practical application quantified with a slaughterhouse wastewater [59]. Similar to rice husk char [60], the leftover solid portion of CH after AD is a good soil supplement, thus further minimizing the organic waste disposal challenge while contributing to clean energy security.

Conclusion

Methane production and organic solid reduction rate from cowpea hull co-digested with waste-activated sludge were enhanced by thermal pretreatment and N_2-nanobubble water addition, increasing methane production by 12.9 and 20.9%, respectively, compared to fresh cowpea hull. However, no significant effect was observed with Air-NBW. This result can be attributed to the improved substrate's digestibility and deactivation of antinutrients, which facilitated the F/M interaction rate. The suppression of trypsin inhibitors may have aided the AD process, but the mechanisms in anaerobic microbial activity need to be clarified. The results reveal that 47.13 to 68.90 million m^3 of biogas generated by co-fermentation of untreated and thermally treated cowpea hull with WAS can be converted to 75.36 GWh, which displaces 0.27% of the total fossil fuel-based electricity capacity generated in 2020. Besides, the green energy generated can mitigate methane emissions by recycling organic waste that would otherwise be landfilled.

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Authors' contributions Conceptualization was contributed by CTO, DR, and YH; methodology was contributed by CTO, and DR; data analysis was contributed by CTO, and DR; first draft was contributed by CTO, and DR was contributed by review and editing, CTO, DR, and YH; supervision was contributed by YH

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Availability of data and materials All available datasets are reported in this work.

Declarations

Conflict of interest The authors declare no conflict of interest.

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